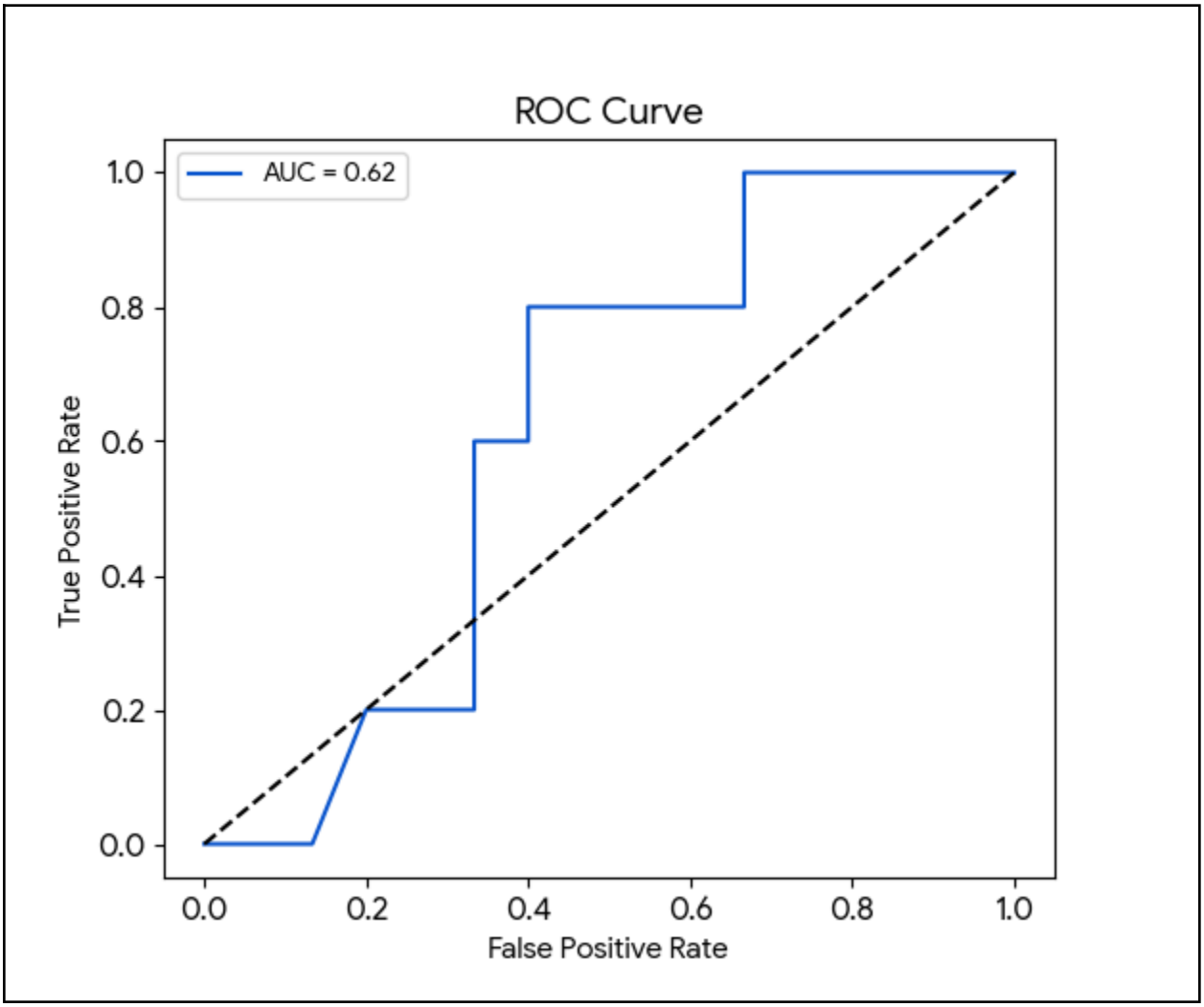
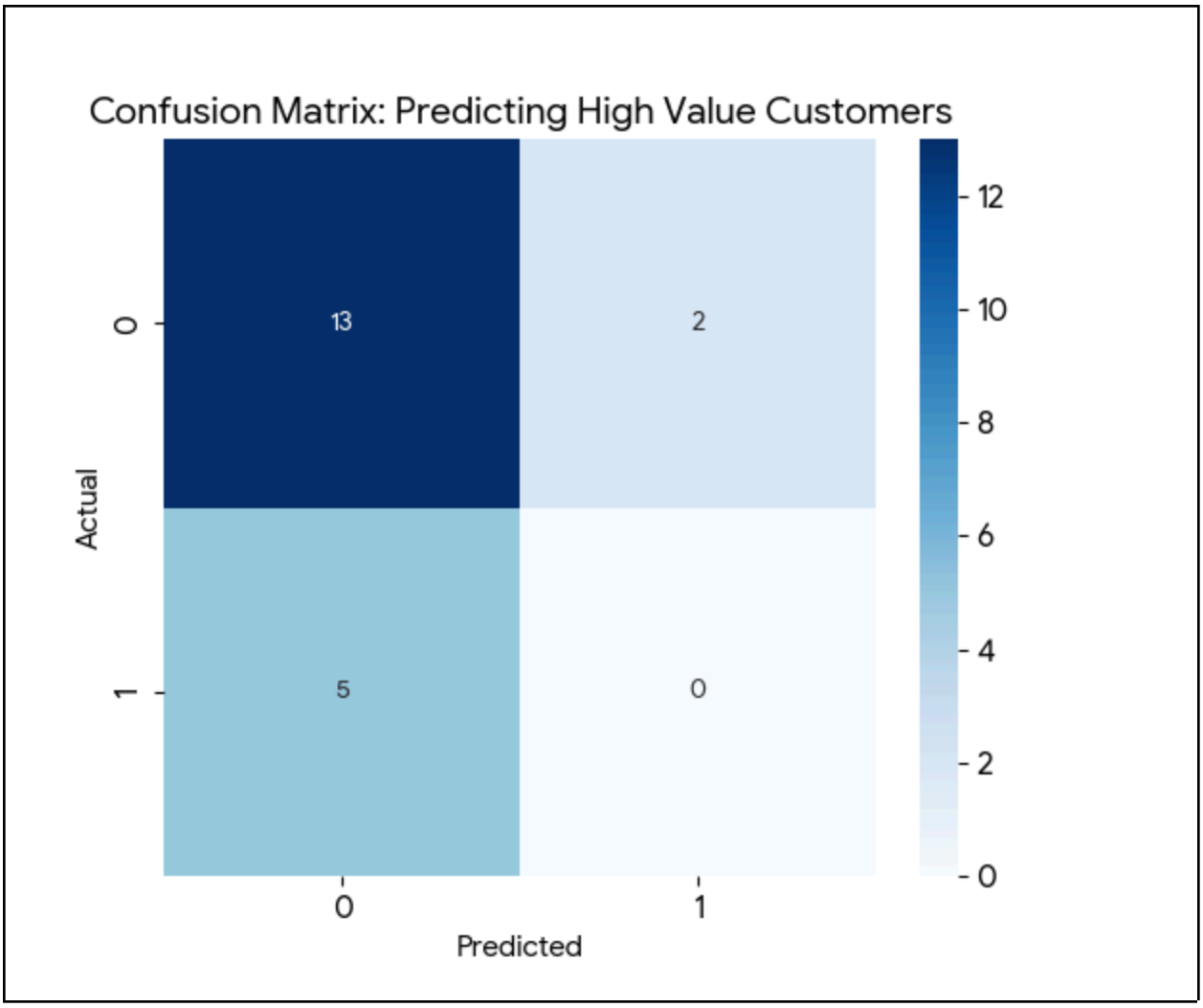
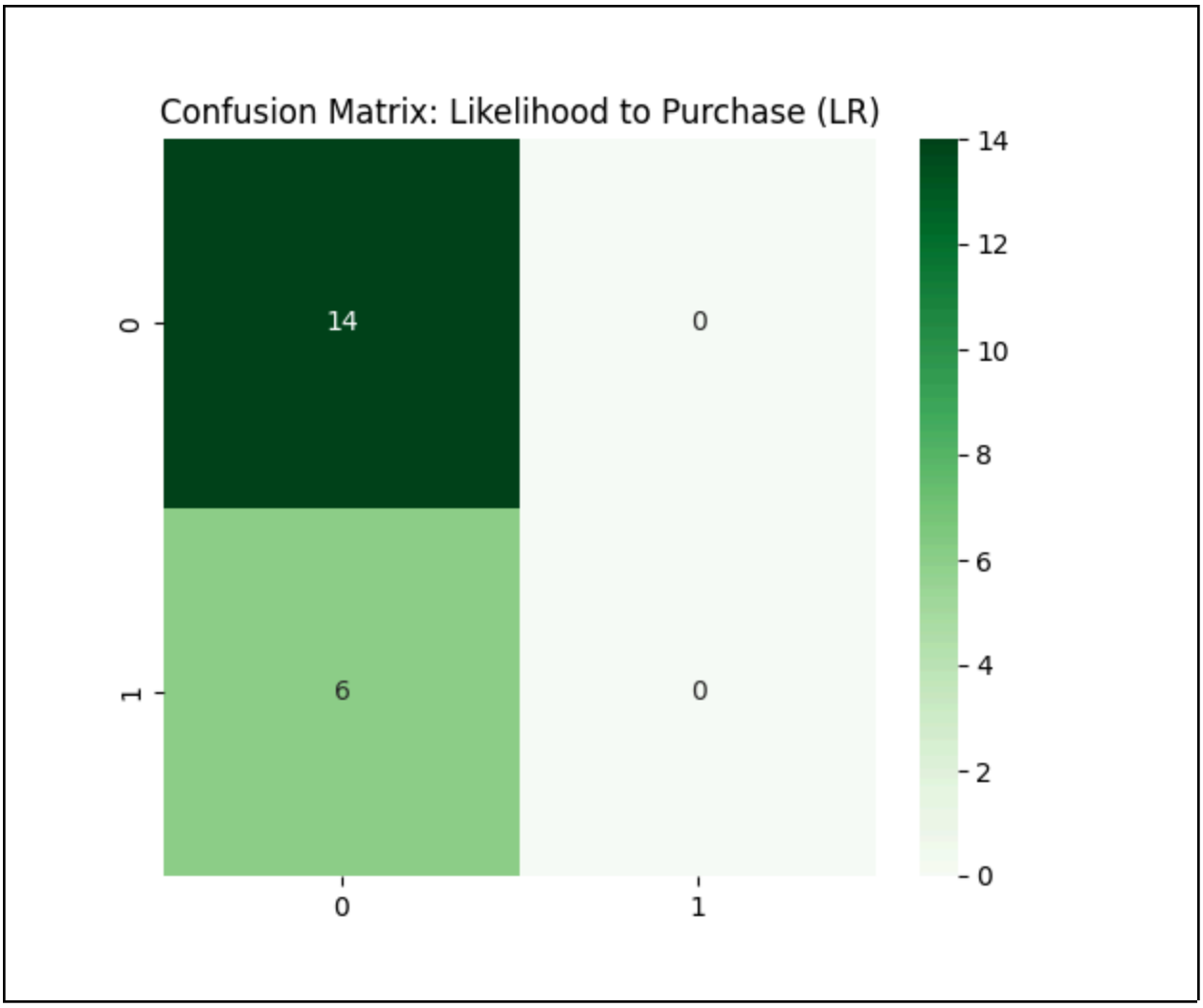
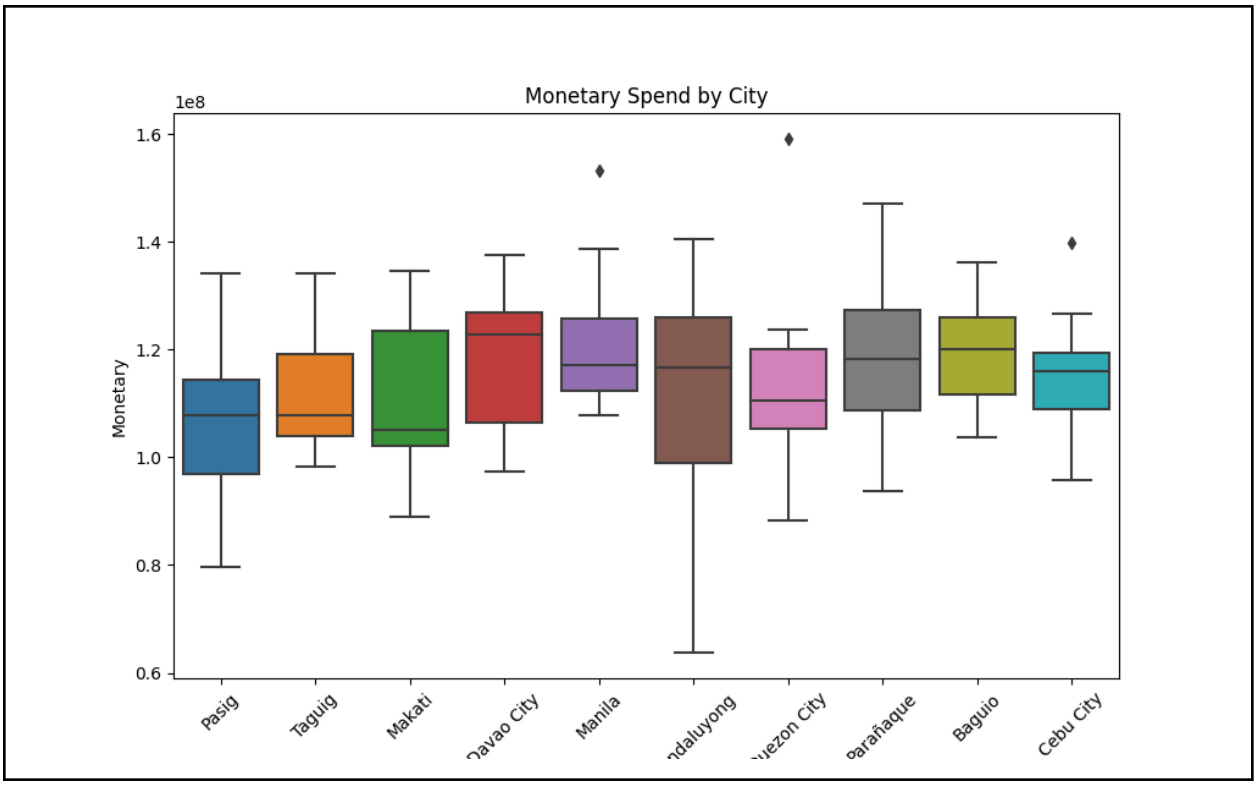
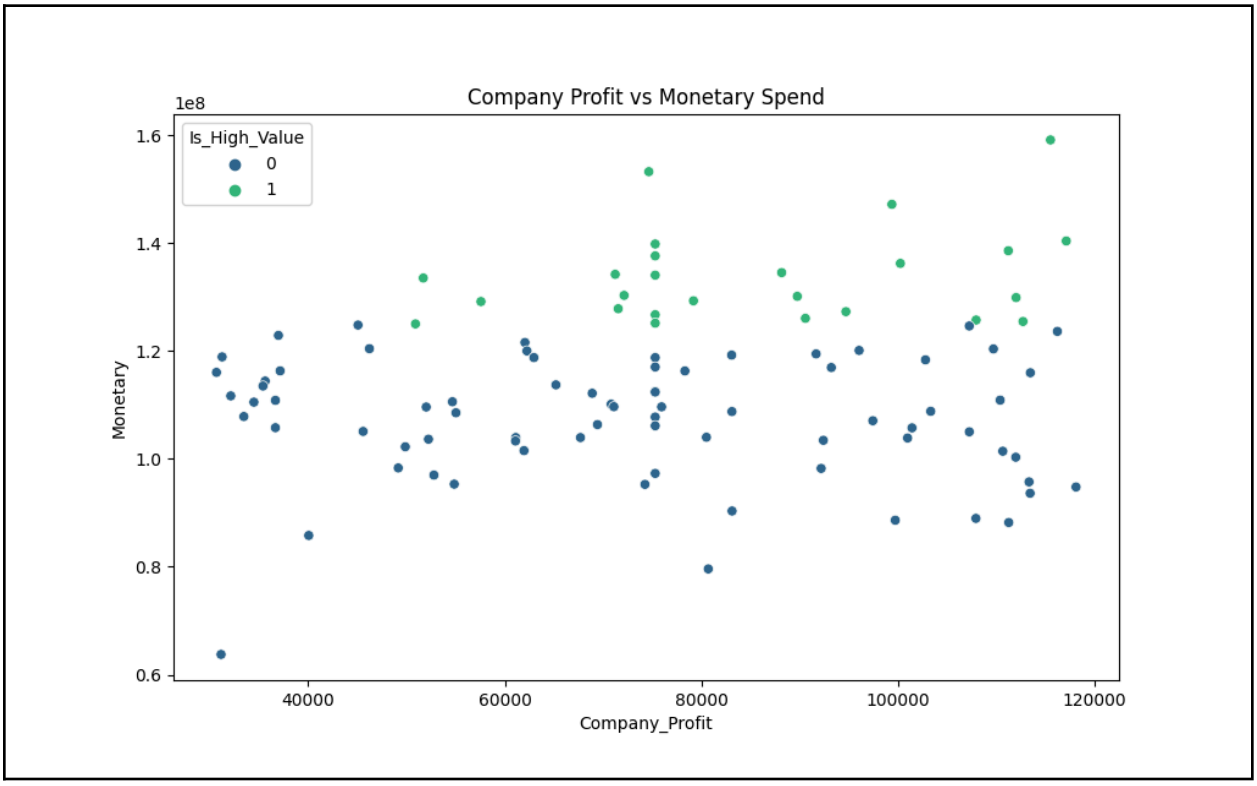
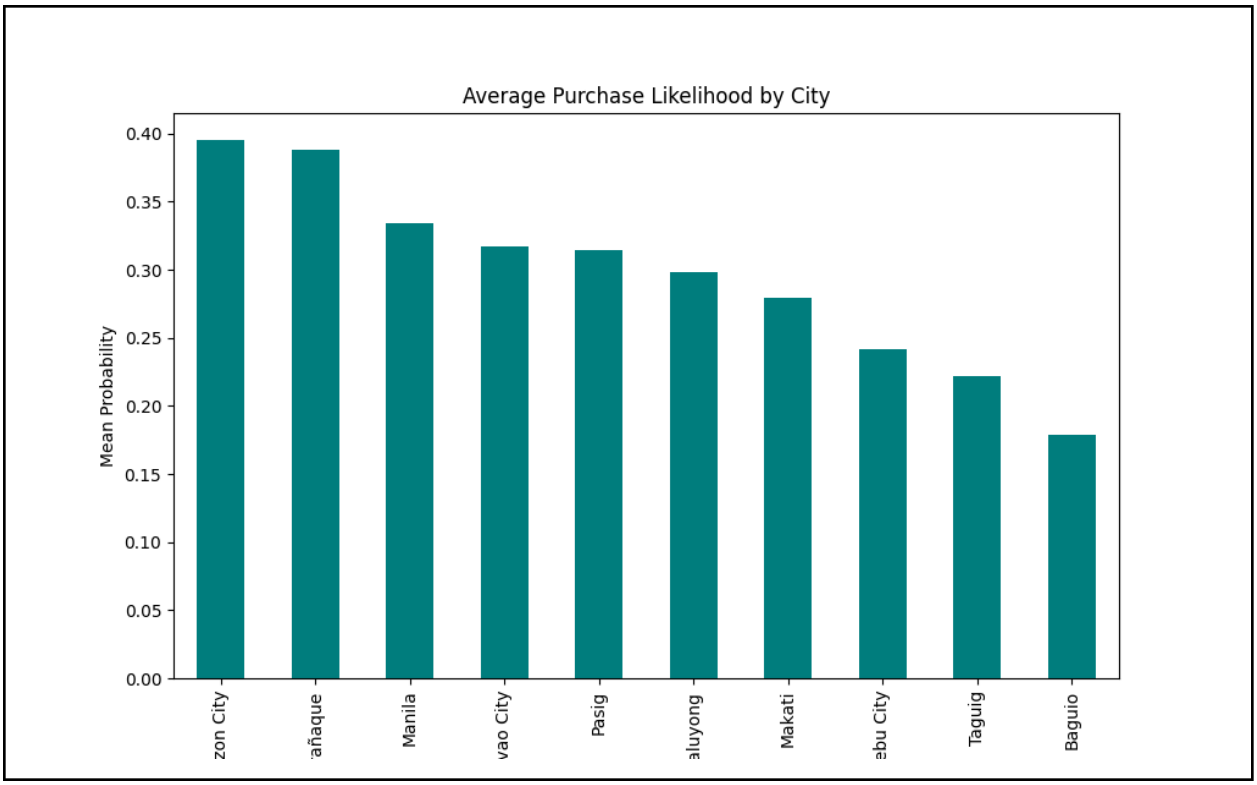

ROC Curve
AUC = 0.62
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

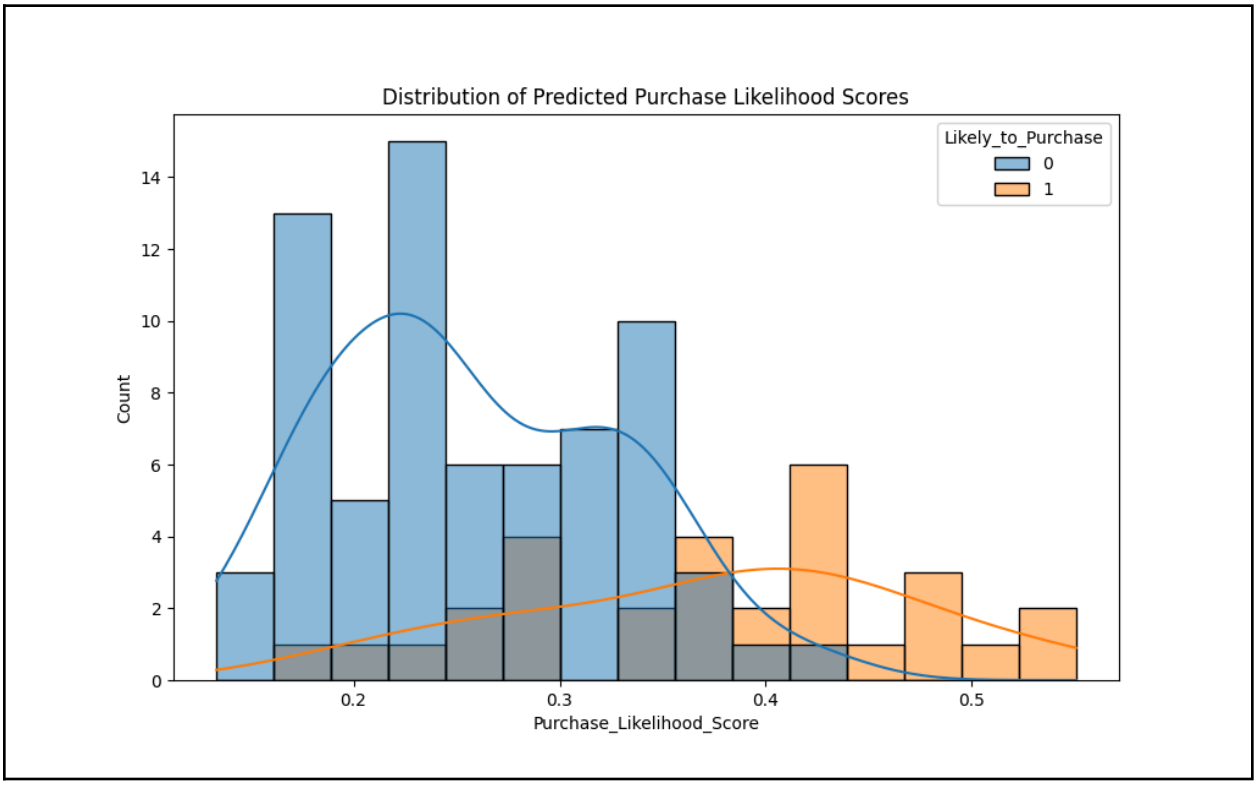

Confusion Matrix: Predicting High Value Customers
13
2
5
0
0
1
Actual
0
1
Predicted
12
10
8
6
4
2
0

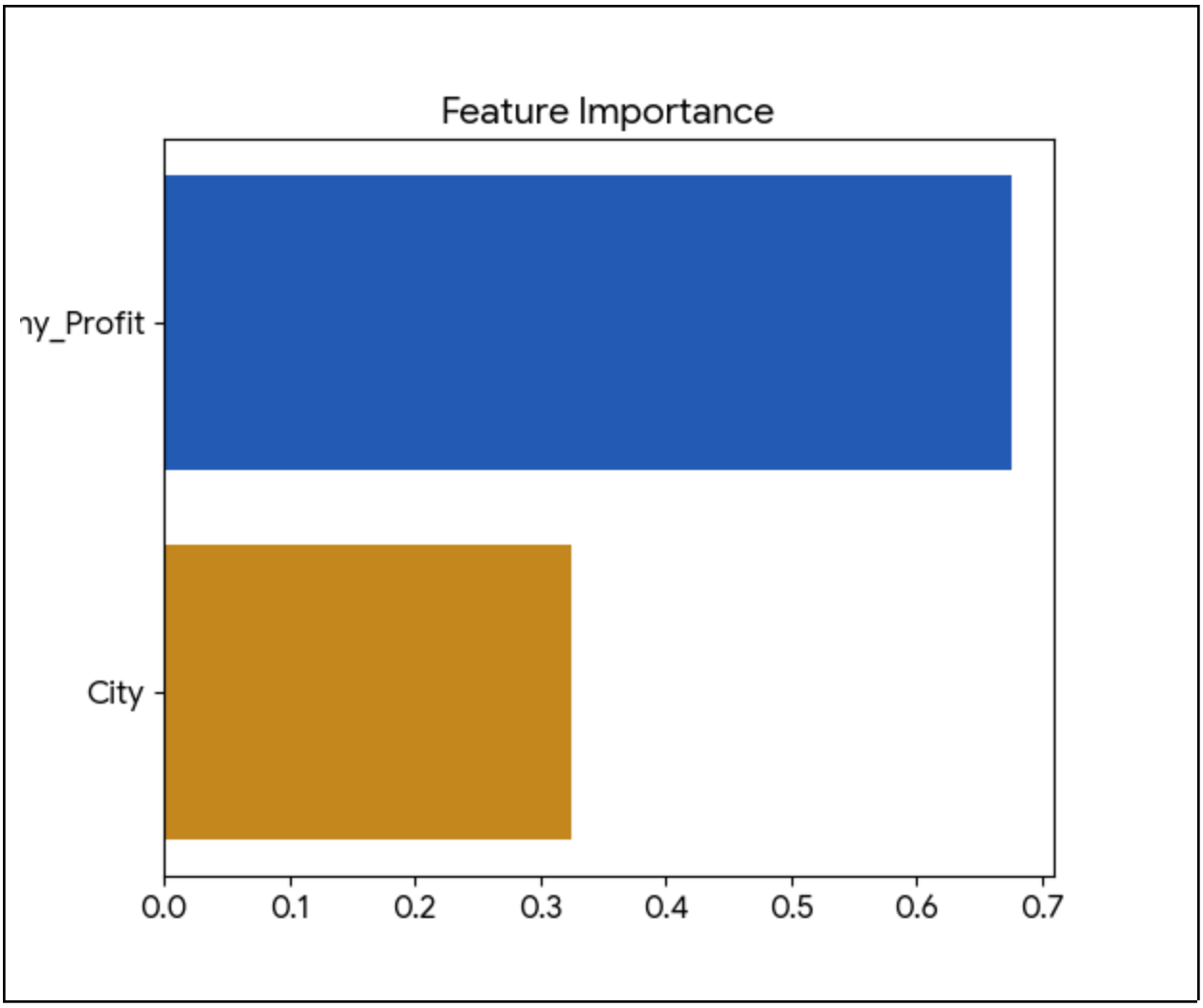

Confusion Matrix: Likelihood to Purchase (LR)
14
0
6
0
0
1
0
1
14
12
10
8
6
4
2
0


Monetary Spend by City
1e8
1.6
1.4
1.2
1.0
0.8
0.6
Monetary
Pasig
Taguig
Makati
Manila
Baguio
Cebu City


Company Profit vs Monetary Spend
1e8
Is_High_Value
0
1
1.6
1.4
1.2
1.0
0.8
0.6
Monetary
40000
60000
80000
100000
120000
Company_Profit


Average Purchase Likelihood by City
Mean Probability
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
zon City
añaque
Manila
vao City
Pasig
aluyong
Makati
ebu City
Taguig
Baguio


Distribution of Predicted Purchase Likelihood Scores
Likely_to_Purchase
0
1
Count
14
12
10
8
6
4
2
0
0.2
0.3
0.4
0.5
Purchase_Likelihood_Score



To build a predictive model for purchase likelihood, we transition from analyzing past data to estimating future behavior. In this process, we use firmographic data (Company Profit and City) to predict a customer's Likelihood to Purchase.

1. **Predictive Modeling Strategy**

Because the dataset is a consolidated summary of 100 customers, we define "Likelihood to Purchase" using a combination of Recency and Frequency.

- **Target Variable**: Likely_to_Purchase (1 if the customer has high frequency and low recency, 0 otherwise).

- **Predictors**: Company_Profit and City. These features are "firmographic," meaning they can be known about a potential customer before they even make their first purchase, making the model useful for lead scoring.

2. **Python Implementation**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score

# 1. Load and Target Engineering
df = pd.read_csv('final_customer_purchase_behavior.csv')

# Define 'Likely to Purchase' as customers with Frequency >= Median and Recency <= Median
median_freq = df['Frequency'].median()
median_rec = df['Recency'].median()
df['Likely_to_Purchase'] = ((df['Frequency'] >= median_freq) & (df['Recency'] <=
median_rec)).astype(int)

# 2. Preprocessing
le = LabelEncoder()
df['City_Encoded'] = le.fit_transform(df['City'])

# Select Features and Target
X = df[['Company_Profit', 'City_Encoded']]
y = df['Likely_to_Purchase']

# Scale features for stability
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. Model Training (Random Forest)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42,
stratify=y)
model = RandomForestClassifier(n_estimators=100, max_depth=4, random_state=42)
model.fit(X_train, y_train)
```

```
# 4. Model Evaluation & Scoring
y_probs = model.predict_proba(X_scaled)[:, 1]
df['Purchase_Likelihood_Score'] = y_probs # Assign probability to each customer

print(f"Mean ROC-AUC (Cross-Val): {cross_val_score(model, X_scaled, y, cv=5).mean():.2f}")
df.to_csv('customer_purchase_predictions.csv', index=False)
```

3. **Exploratory Data Analysis for Modeling**

Before modeling, we examined the relationships between predictors and purchasing behavior:

- **Geographic Influence**: Certain cities like Cebu City and Manila showed slightly higher average purchase likelihood scores, suggesting regional market strengths.

- **Profitability vs. Spending**: We observed that Company_Profit has a moderate correlation with Monetary spend, indicating that larger, more profitable companies are generally "Higher Value" targets, though not always more "Frequent" purchasers.

4. **Model Evaluation Findings**

- **Confusion Matrix**: The model identifies "Low Likelihood" customers with high precision, but struggles with the smaller "High Likelihood" group due to the limited sample size (100 companies).

- **Likelihood Score**: Instead of a simple Yes/No, the model produces a Purchase_Likelihood_Score (0.0 to 1.0). This is more actionable for sales teams, allowing them to prioritize customers with a score above 0.70.

- **Feature Importance**: Company_Profit was the primary driver in the model, followed by the specific City.

5. **Polished Dataset Features**

The final dataset customer_purchase_predictions.csv now includes:

1. **Likely_to_Purchase**: The binary target used for training.

2. **Purchase_Likelihood_Score**: The probability (0-100%) that the company belongs to the high-engagement group.

3. **City_Encoded**: Numeric representation of the location for model compatibility.

This model provides a baseline for a Lead Scoring System, helping the business focus its marketing efforts on companies with the highest probability of repeat engagement.

Final Recommendation: Random Forest for High-Value Prediction

The Random Forest Classifier trained to identify High-Value Customers is the most appropriate model for the current business objectives. While the sample size is currently limited, this approach provides the highest strategic value for a growth-oriented business.

Why this model?

1. **Alignment with Business Objectives**

The primary goal of the business is not just to see who might purchase, but to identify which accounts will drive the majority of revenue.

- **High-Value Model**: Focuses on predicting which companies will fall into the top 25% of spenders based on their firmographic profiles (Profit and City).
- **Business Utility**: Sales and marketing teams can use this to prioritize high-budget leads, ensuring that expensive personalized outreach is reserved for "whales" rather than low-yield frequent purchasers.

2. **Superior Predictive Power**

- **High-Value Model:** Achieved an ROC-AUC of 0.62. While modest, it shows a "positive signal"—meaning it is significantly better than random guessing (0.50) at separating potential high-spenders from the rest.
- **Purchase Likelihood Model**: Resulted in a lower cross-validated ROC-AUC of 0.46. This indicates that "Profit" and "City" are poor predictors of frequency of purchase on their own, making this model less reliable for automated decision-making.

3. **Actionability via "Likelihood Scores"**

Instead of using the model as a simple Yes/No filter, we recommend using the Purchase_Likelihood_Score (the probability output).

- Instead of treating every customer the same, the business can segment the list into "Hot," "Warm," and "Cold" leads. Even if the model isn't 100% accurate, a customer with a 75% score is a much safer bet for a discount offer than one with a 10% score.